

Loads, Not Bequests: An Order-Independent Decomposition of the Annuity Puzzle

Replication package for Tharp (working paper). A calibrated lifecycle model nests the channels proposed to explain low voluntary annuity demand among US retirees in a single framework and attributes the annuity-demand gap with an exact, order-independent Shapley decomposition. The channels are organized in three layers:

- **Seven rational channels:** pre-existing Social Security annuitization, bequest motives, combined medical-expenditure and health-mortality (Reichling-Smetters) risk, subjective survival pessimism, pricing loads, inflation erosion, and public-care aversion (a long-term-care/Medicaid channel). Medical risk and the R-S correlation enter as a single channel because the correlation has quantitative bite only through medical risk.
- **Two preference channels:** age-varying consumption needs (Aguiar-Hurst 2013) and state-dependent utility (Finkelstein-Luttmer- Notowidigdo 2013).
- **Two exploratory behavioral channels:** source-dependent utility (Blanchett-Finke 2024, 2025) raises annuitization by converting portfolio wealth into “spendable income”; a narrow-framing purchase-event disutility (Hu-Scott 2007; Brown 2008; Chalmers-Reuter 2012) suppresses it through loss aversion over the unrecouped premium until breakeven. Behavioral parameters ($\lambda_W = 0.625$, $\psi = 0.05$) are treated as exploratory literature-magnitude best guesses, not moment-matched estimates.

Headline result

The contribution is the **order-independent ranking** of demand channels, not a point prediction of the ownership level. An exact Shapley decomposition over all 512 subsets of the nine structural channels (seven rational + two preference, with public-care aversion entering as a structural-LTC channel) attributes the demand gap without dependence on the order in which channels are introduced. Pricing loads are the dominant suppressor (~81% of the structural drop), with survival pessimism and the combined medical-expenditure/health-mortality channel as the co-leading second tier, and bequest motives – the most-cited explanation in the literature – mid-pack.

The ranking is stable where the predicted **level** is not: predicted ownership is knife-edged in risk aversion (it spans 0% to roughly 22% across γ in [2.0, 3.0]) because the extensive margin sits on a fixed-cost threshold interacting with the pricing load, while the channel ranking holds across that risk-aversion range, across wealth bands, and at alternative baseline money’s worth ratios (boundary cases disclosed in the text). The nine-channel structural baseline predicts roughly 8% ownership under the baseline calibration ($\gamma = 2.5$, $MWR = 0.87$, Wettstein-2021 modern pricing); predicted ownership rises with wealth and concentrates in the top band, with near-zero predicted demand below it, because the pricing load makes immediate annuitization value-destroying for all but the wealthiest. The observed HRS gradient also rises with wealth but is interior in the middle of the distribution, where the single-product model predicts none. An exploratory eleven-channel Shapley over 2,048 subsets adds the two behavioral parameters; they carry the largest absolute contributions essentially by construction and reorder the structural attributions, which is why the ranking is read off the nine-channel game.

Two HRS measures of US lifetime annuity ownership (computed on the same wealth-restricted analysis sample as the model predictions) are reported in parallel as out-of-sample checks, neither entering the calibration: 3.64% (95% CI [2.95%, 4.49%], the cleaner fat-file q286 lifetime annuity

contract indicator) and 6.06% (95% CI [5.15%, 7.11%], the conventional any-annuity income proxy used in prior literature).

All headline numbers in the manuscript are auto-generated from analysis CSV outputs by `scripts/export_manuscript_numbers.jl` and locked by `test/test_manuscript_numbers.jl`. Don't hand-edit `paper/numbers.tex` or the table inputs.

Requirements

- **Julia:** 1.12+ (tested on 1.12.5)
- **Key packages:** `Interpolations.jl`, `Distributions.jl`, `Optim.jl`, `QuantEcon.jl`, `DataFrames.jl`, `CSV.jl`, `Plots.jl`
- **Hardware:** 4 GB RAM minimum; 16 GB recommended for parameter sweeps

All dependencies are pinned in `Manifest.toml`.

Quick Start

```
julia --project=. -e 'using Pkg; Pkg.instantiate()'  
julia --project=. run_all.jl
```

`run_all.jl` runs the test suite first by default; pass `--skip-tests` for faster iteration during development (NEVER for production). The compute stages require the HRS population extract, which is not redistributable and must be regenerated once from a free RAND HRS download – see “Data availability” below. The lock-test suite and the fast verification path run without it.

To run only the tests, use `julia --project=. test/runtests.jl` (the canonical test runner). The project is intentionally not configured as a package for `Pkg.test()`: each test file is included in a separate process to avoid the module re-definition conflicts that the codebase’s include-based loading would otherwise produce.

Runtime

The full pipeline takes approximately 10 hours on a 192-vCPU AWS c7a.48xlarge spot instance, or 12+ hours on a 16-core Mac Studio. Individual scripts run in minutes to tens of minutes. Use `julia --project=. -p N` to parallelize compute-heavy scripts across N cores.

Fast verification path

For a laptop-scale check without the full 10-hour run: the test suite (`julia --project=. test/runtests.jl`, ~6 minutes) locks every manuscript number to its source CSV and exercises the solver on production-relevant configurations; `EMG_SMOKE=1` runs the extensive-margin gate on a coarse grid in minutes; and any single exhibit script can be run alone (see the exhibit map below). The committed CSVs are the outputs of a from-scratch production run, so the locks verify the deposit without re-solving.

For AWS execution see `scripts/aws/launch.sh` (provisions a spot instance, syncs the project, runs the full pipeline, and auto-terminates on completion).

Configuration

Baseline parameters are defined in `scripts/config.jl`. Key values: `gamma=2.5`, `beta=0.97`, **`MWR=0.87`** (a deliberately conservative modern-market choice above the Mitchell-et-al./Wettstein anchors), `inflation=2%`, DFJ bequests (`theta=56.96`, `kappa=$272,628`), hazard multipliers `[0.50, 1.0, 3.75]`, survival pessimism `psi=0.96` (Heimer-Myrseth-Schoenle 2019; Payne et al. 2013), age-varying needs `delta_c=0.02`, health-utility weights `[1.0, 0.92, 0.82]` (FLN 2013 central midpoint), **`lambda_W=0.625`** (source-dependent utility, Blanchett-Finke 2024/2025 spending differential point estimate; exploratory), **`psi_purchase=0.05`** (purchase-event disutility, literature-magnitude best guess; exploratory, not moment-matched), 9-node Gauss-Hermite quadrature, 80x30x101 production grid (`W x A x alpha`).

Data availability

The model is solved from calibration parameters and the processed inputs below; no proprietary data are required to reproduce the structural results, tables, or figures. All raw data used for calibration come from the Health and Retirement Study (HRS), publicly available at no cost to registered users (<https://hrsdata.isr.umich.edu>).

The repository ships with all *aggregate* processed CSVs (transition matrices, hazard ratios, income profiles, band-level counts). Two person-level extracts are NOT shipped because HRS conditions of use restrict redistribution of individual-level records: `lockwood_hrs_sample.csv` (the HRS population file over which the model is evaluated) and `hrs_validation_sample.csv`. The compute stages (Stage 1 onward) require the population file, so running `run_all.jl` end to end requires regenerating it first from a free RAND HRS download (`ANNUITY_FORCE_HRS_REBUILD=1`; see below). Without HRS access, the committed table/figure CSVs and the lock-test suite still verify the deposit (see “Fast verification path”). HRS conditions of use prohibit redistributing the person-level extracts to anyone, including journals; the deposit instead provides the generating programs and reconstruction documentation, and any journal replication requirement is handled per the HRS journal replication policy (data exemption request, with reviewers obtaining the raw files directly from HRS under their own free registration).

Processed CSVs. Each is regenerated by one calibration script from the raw RAND HRS inputs. All aggregate CSVs are checked in; the two person-level extracts (`lockwood_hrs_sample.csv`, `hrs_validation_sample.csv`) are NOT shipped (HRS conditions of use) and must be regenerated locally:

- `data/processed/lockwood_hrs_sample.csv` – HRS population sample (4,258 person-wave observations, single retirees aged 65–69, waves 5–9; person-level, not shipped); `calibration/build_hrs_sample.jl`.
- `data/processed/hrs_lifetime_ownership.csv` – HRS lifetime annuity contract indicator (q286 fat-file series), pooled across waves 5–9; `calibration/compute_lifetime_ownership_rate.jl`.
- `data/processed/ss_income_profile.csv` – claimer-conditional Social Security and pension-only DB income by wealth band (2014 dollars); `calibration/build_ss_profile.jl`.
- `data/processed/health_transitions_age_bands.csv` – three-state health Markov transition matrices by age band; `calibration/estimate_health_transitions.jl`.
- `data/processed/hrs_hazard_ratios.csv` – health-specific mortality hazard ratios by age band; `calibration/compute_hazard_ratios.jl`.
- `data/processed/hrs_lifetime_ownership_by_band.csv` – lifetime annuity ownership by wealth band, person-wave and person-level counts; `calibration/q286_by_wealth.jl`.
- `data/processed/group_access_by_band.csv` – group-annuity access rates by wealth band (employer-pension linkage); `calibration/build_group_access.jl`.
- `data/processed/hrs_validation_sample.csv` – person-level cross-sectional extract for the empirical ownership-gradient checks (NOT shipped; re-

generate via `ANNUITY_FORCE_HRS_REBUILD=1`; `calibration/build_validation_sample.jl`.
- `data/processed/hrs_acquisition_decomposition.csv` and `hrs_acquisition_decomposition_robust.csv`
– HRS annuity-acquisition covariate decomposition (baseline and robustness); `calibration/decompose_hrs_acquisition.jl`
and `..._robust.jl`.

Raw inputs (not checked in; institutional access required): - RAND HRS Longitudinal File 2022 (v1) – free, requires HRS account, <https://hrsdata.isr.umich.edu>. - HRS public survey “fat files” – pension/wealth modules per wave.

See `data/raw/README.md` for full provenance and expected paths.

The pipeline gracefully skips raw-data stages if the source files are absent and falls back to the checked-in processed CSVs. To force full regeneration when raw inputs are present, set `ANNUITY_FORCE_HRS_REBUILD=1`.

Redistribution of HRS-derived files. Two intermediate CSVs (`lockwood_hrs_sample.csv`, `hrs_validation_sample.csv`) hold individual person-wave records extracted from the RAND HRS file. HRS data are distributed under a conditions-of-use agreement that restricts redistribution of individual-level records, so these two files are NOT included in this repository (they are gitignored and absent from its history). Users regenerate them from their own RAND HRS download via `ANNUITY_FORCE_HRS_REBUILD=1`; see the Data availability section for the HRS access route. All checked-in processed CSVs are aggregates (transition matrices, hazard ratios, income profiles, band-level ownership counts) and carry no individual-level records; these and all calibration targets may be shared freely.

Reproducing the exhibits

`run_all.jl` regenerates every table and figure from the processed inputs. To rebuild one exhibit, run its script directly (each writes to `tables/` and `figures/`); `scripts/export_manuscript_numbers.jl` then refreshes `paper/numbers.tex`, and `scripts/validate_pipeline.jl` checks that every manuscript macro is defined and matches its source CSV.

Manuscript exhibit	Generating script
Figures 1–5	<code>scripts/generate_figures.jl</code>
Shapley decomposition (9- and 11-channel)	<code>scripts/run_subset_enumeration.jl</code>
Pairwise interactions, retention path	<code>scripts/run_decomposition.jl</code>
Welfare CEV grid	<code>scripts/run_welfare_analysis.jl</code>
CEV / welfare counterfactuals	<code>scripts/run_welfare_counterfactuals.jl</code>
Social Security cut robustness	<code>scripts/run_ss_robustness.jl</code>
gamma / inflation robustness	<code>scripts/run_robustness.jl</code>
Implied risk aversion	<code>scripts/run_implied_gamma.jl</code>
Moment validation	<code>scripts/run_moment_validation.jl</code>
Empirical ownership gradients	<code>scripts/run_empirical_validation.jl</code>
Pashchenko comparison	<code>scripts/run_pashchenko_comparison.jl</code>
DIA comparison	<code>scripts/run_dia_analysis.jl</code>
gamma-stability of the ranking	<code>scripts/run_shapley_gamma_stability.jl</code>
Monte Carlo uncertainty	<code>scripts/run_monte_carlo_uncertainty.jl</code>
Extensive-margin gate (F^* distribution, smoothed gradient)	<code>scripts/run_extensive_margin_gate.jl</code>

Manuscript exhibit	Generating script
Gate robustness (“killer table”)	<code>scripts/run_gate_robustness.jl</code>
Band value-destruction diagnostic	<code>scripts/run_band3_diagnostic.jl</code> + <code>scripts/emit_band_value_destruction_table.jl</code>
Model-vs-data wealth-band table	<code>scripts/emit_model_vs_data_band_table.jl</code>
Channel-partition robustness	<code>scripts/run_partition_robustness.jl</code>
psi=0.981 focal-value ranking	<code>scripts/run_psi981_shapley.jl</code>
Grid-robustness of the ranking	<code>scripts/run_grid_robustness_shapley.jl</code>
Seven-channel sub-game	<code>scripts/compute_subgame_shapley.jl</code>
Alternative-baseline Shapley + by-band policy	<code>scripts/run_referee_proofing.jl</code>
MWRs	
Grid convergence, Euler residuals	<code>scripts/grid_convergence_full.jl</code> , <code>scripts/run_euler_diagnostics.jl</code> -> <code>scripts/emit_diagnostic_tables.jl</code>

Development vs. production grids. Grid sizes and quadrature nodes are set in `scripts/config.jl`. Production exhibits use the 80x30x101 grid (W x A x alpha) with 9-node quadrature. For fast iteration, coarsen the grid in `config.jl` (e.g. 40x15x51): the channel ranking is stable to grid coarsening, but the predicted ownership *level* is not, and levels should be quoted only from a production run.

On the ownership level. Predicted ownership near 6–8% is a step function of the threshold wealth band whose marginal household flips in or out of participation, so intermediate specifications can differ by a few tenths of a percentage point across solution paths (for example, the eight-channel decomposition step and the `chi_LTC` robustness cell bracket the same region without coinciding). These intermediate levels are not interchangeable and should not be quoted against one another. The headline nine-channel level (7.9%) is stable across grid and quadrature refinement; the paper’s claims rest on the channel *ranking*, which is order-independent and stable across risk aversion.

Code Organization

<code>src/</code>	Model source code (module: <code>AnnuityPuzzle</code>)
<code>scripts/</code>	Analysis scripts (decomposition, welfare, robustness)
<code>scripts/config.jl</code>	Canonical parameter configuration
<code>test/</code>	Test suite (including manuscript-number lock)
<code>data/processed/</code>	Processed calibration inputs
<code>tables/</code>	LaTeX and CSV output tables
<code>figures/</code>	Publication-quality figures (PDF and PNG)
<code>paper/</code>	Manuscript and appendix
<code>run_all.jl</code>	Master pipeline script

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